

Report prepared at the request of Chris Coad, Director of Business Services
Brecksville-Broadview Hts. City School District

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March 28, 2019

OSBA was retained to evaluate impacts on the Brecksville-Broadview Heights school district based upon the anticipated closure of Central School. This evaluation was requested to address four specific points. It does not consider any costs or savings associated with building level or educational infrastructure changes.

- 1) Review the estimated routing impacts to move students to middle school routes and to elementary routes.
- 2) Review the estimated need for additional buses to accommodate additional students on MS and Elementary tiers.
- 3) Forecast the economic impact of eliminating the Central routes and adding additional routes to MS and Elementary tiers
- 4) Review the traffic patterns at the MS/HS

1) Review the estimated routing impacts to move students to middle school routes and elementary routes.

Currently the district transports 408 students to the Central campus on 20 buses. (Data from district routing reports.) These students are in fourth and fifth grades. The plan calls for students in fifth grade to be relocated to the middle school building and students in fourth grade to be relocated to district elementary buildings.

The district currently operates a four tier routing plan, with Central occupying the third tier in the plan. The displacement of these students to the Middle school (second tier) and Elementary schools (fourth tier) impacts on the total routing structure of the district. We can immediately eliminate the 20 trips being run on the third tier, which decreases direct operating expenses of those vehicles. The fifth grade students will be picked up on the existing Middle school routes and fourth grade students will be picked up on the appropriate elementary school routes.

Both the middle school and elementary school routes have to be evaluated to determine if there is sufficient seating available to add additional students. It is reasonable to assume that some of these students may require additional bus stops that are not scheduled on the existing

routes. The net impact on the routes is that they will become slightly longer-both due to additional students loading and extra stops if required.

We have to consider the existing seating capacity and determine if additional buses will need to be assigned to those programs. There is a logistical trade-off between seating capacity and time constraints. With unlimited time constraints we can increase the carrying capacity of the existing buses, thereby limiting the need to schedule additional buses into the fleet plan. Conversely, if routing time is limited, the capacity of the buses is limited not by seats but by routing time, and we may need to add additional buses to carry the additional students.

2) Review the estimated need for additional buses to accommodate additional students on MS and Elementary tiers.

The first requirement in this element is to evaluate the ridership of the newly configured fourth and fifth grades. Transportation staff in the district expect a higher number of riders at both grade levels due to the relocation of educational services and loss of after school care. Based on existing data, BBH staff suggested using a ridership rate of 81% of eligible students. Based upon OSBA's experience with other districts, this is a reasonable basis for ridership estimates.

Using district enrollment data, the ridership is forecast as follows:

Grade level	ADM	Ridership percentage	Forecast riders
Fourth grade	273	81%	221
Fifth grade	289	81%	234

The data indicate a slightly higher projected ridership than currently at Central but is within expectations and reasonable.

Impact on Middle school routes:

Currently there are 711 students riding on 27 buses, with an average ridership of 27 students per bus. Routes range from 20 to 40 minutes in length, both in the morning and afternoon. Current ridership is low when compared with vehicle capacity. The majority of the buses used by the district are rated at 71 passengers, with a realistic capacity of 47 passengers per bus (assuming 2 riders per seat). For a more effective use of buses, which ultimately reduces transportation cost, the routing goal should be to increase the ridership on the buses. With the introduction of additional riders into this routing tier, the district has the opportunity to move toward this goal. The mitigating factor will be the amount of route time available.

If no bell time changes are made, or if the Middle school routes are move as much as 15 minutes later in the day there are no obvious time conflicts that would preclude the district from operating the buses more efficiently.

ASSESSMENT OF ROUTING IMPACT BY CENTRAL SCHOOL CLOSURE

If we maintain current ridership of 26 students per bus, we need to add 10 buses to this route time. If use a more realistic goal of 35 riders per bus, we can transport this new total group of riders with 29 buses, which is only two more than currently scheduled. In a third option, we add 6 buses to this tier resulting in an average of 30 riders per bus.

Middle school	Riders	Buses	Riders / bus
Current	711	27	26.3
Incoming 5 th grade	273		
Option 1	984	27	36
Option 2	984	29	34
Option 3	984	33	30

Impact on Elementary school routes:

There are three elementary schools, with a total ridership of 777 students on 22 buses. The average ridership is 35 students per bus. Each elementary is different, with Highland at 27 riders per bus, Hilton at 41 riders per bus, and Chippewa at 35 riders per bus.

We have 221 new riders between these schools. Assuming a proportional distribution and maintaining the current ridership ratio we would need 6 additional buses. If we increase the ridership per bus we can decrease the number of additional bus trips required. A typical routing goal for elementary students is 45 passengers per bus, which would require 22 buses. This is exactly the same number of buses as currently used. With the increased students it would be reasonable to assume a moderate growth in the number of buses. If we add one vehicle into each building, we would need to add 3 trips in the afternoon and would achieve a ridership of 39 passengers per bus.

Elementary	Riders	Buses	Riders/bus
Current	777	22	35
Incoming 4 th grade	221		
Option 1	998	22	45
Option 2	998	25	39

3) Forecast the economic impact of eliminating the Central routes and adding additional routes to MS and Elementary tiers

Performing a cost analysis of these changes is dependent upon both the number of trips that are added to each of the other tiers, as well as potential changes in the school bell schedules.

To develop a cost basis, we use the data reported last year to ODE on the T1 and T2 transportation reports. Analysis of these reports shows a total operating cost per bus of \$60,063.

When we evaluate the costs to add additional buses, or to adjust tier assignments the cost impact is less than the total cost. Support and overhead costs do not change in these adjustments. The operational cost to add an additional vehicle to the fleet is calculated at \$52,261 per bus. The cost of eliminating one trip from an existing bus route in the middle of a route set is calculated at \$1909 per bus. There is no labor cost impact because the route is in the middle of a route set and will not affect the driver's total pay time. Likewise, adding a tier trip to an existing bus is similarly calculated at \$1909 per bus, with no labor impact.

If all school times are left as currently schedule, we consider three scenarios.

- 1) Eliminate Tier 3, absorb 5th grade riders into existing MS routes, absorb 4th grade riders into the Elementary routes. This plan saves 20 tier trips, adds no new trips, and results in a net cost savings of \$38,180.
- 2) Eliminate Tier 3, add 2 buses to the existing MS routes to achieve a better ridership ratio, and add 3 buses to the elementary routes to relieve additional routing requirements due to new riders. Evaluating the current fleet utilization schedule, there are 2 unassigned buses available at the MS time shift and there are 7 unassigned buses available at the Elementary time shift, so no vehicle need to be added to the fleet schedule. This plan saves 20 tier trips on the 3rd tier, and adds a total of 5 tier trips for MS and Elementary. This results in a net cost savings of \$28,635.
- 3) Eliminate Tier 3, add 6 buses to the existing MS routes to minimize the impact on current ridership ratios, and add 6 buses to the elementary routes. This will require 4 additional fleet buses because there are only 2 unassigned buses available for MS. There is a net savings from tier reduction of \$22,908 but an additional operational cost for the 4 new buses of \$209,040. The net impact to the district is an added cost of \$186,132.

If the district determines that it wants to adjust the bell times of the remaining buildings, it is necessary to consider the amount of time available for routing on each tier. Currently the fleet is challenged with the interval between the high school and the middle school. This results in early dropoffs at the high school and late dropoffs at the middle school. By eliminating the current third tier, there is a good opportunity to open up the bell schedule to provide appropriate routing times between each of the remaining schools. Effective routing will require that 40 minutes be considered the optimal spread between the bell times. This will allow the district to continue to operate with the existing fleet size and realize the ability to make the routing changes identified in plans 1 and 2 above without additional cost. If the routing windows are reduced below the 40 minute window, a more thorough routing analysis will be required to assess the impact on required fleet size and subsequent added operating costs. Analyzing that impact is beyond the scope of this limited assessment.

Review the traffic patterns at the MS/HS

OSBA's Doug Palmer travelled to the district to observe the traffic flow at the high school and middle school.

The special ed buses park between the entrances opposite the parking lot at the north end of the high school building. The students are the first to load and the buses then exit around small island leaving the parking area past the parent pick up traffic. The buses then exit school grounds using the west drive (red traffic light located here).

The regular ed buses pass the special ed buses and park bumper to bumper on the drive that goes around the football stadium. The first bus in line parks just short of the east entrance into the lower parking lot south of the stadium. The last bus parks just south of the top parking lot. Students exit the high school building by the four doorways on the front side and the doorway on the south side.

The regular ed buses then drive around the stadium, between the tennis courts and the parking lot to the middle school. The buses then travel left around the grass island and exit the school grounds onto Mill Street. The school resource officer directs traffic (the traffic light is in flashing mode) to assist, parents, students and the buses access to Mill Street.

The majority of the fleet travels north on Mill Street. There is a substantial traffic back-up on Mill Street between the schools and Wallings Road. This delay is normally 10-12 minutes in the morning and afternoon.

The middle school bus loading in the afternoon is along the entire front side of the building. Buses park bumper to bumper when arriving, some buses do not return in time from the high school route to arrive in time for the dismissal bell. These students wait either in the grassy area by the building or just inside the doorway.

The middle school buses exit school grounds in the same manner as the high school buses and face the same traffic delays on Mill Street.

The district would benefit by considering a more effective student management system for loading the buses, and then releasing the buses before the student and parent vehicles. This would help to minimize the traffic impact. The most significant impact is a result of the traffic on Mill, and is outside of the district's control.

18 cost analysis

Following is an analysis of the current operating cost per bus specific to this district

T2 Reg ed expenses

School District: Brecksville Broadview

FY18 costs	Total Costs	cost / bus
Reg Bus		
Drivers	905,552	25,154
Sub Bus		
Drivers	36,660	1,018
Total Drivers	942,212	26,173
Supervisors	69,348	1,926
Mechanics	123,637	3,434
Mech Helpers	-	-
Secretary	78,445	2,179
Total Support Staff	271,430	7,540
Fringe	572,599	15,906
Total Staff	1,786,241	49,618
Maint/Repair	164,903	4,581
Fuel	148,143	4,115
Tires/Tubes	30,712	853
Bus Insurance	22,836	634
Other	9,464	263
Total Operations	376,058	10,446
Total Cost	2,162,299	60,064

Bus attendents

4 SPED BUSES

buses	36
Total cost / bus	\$ 60,063.86

Additional veh costs

District: Brecksville Broadview

Item	Annual Avg Cost/bus	Cost adjustment	Additional Cost 1 tier
Reg Bus driver	25,154	100%	\$ 25,154.00
Sub bus driver	1,018	100%	\$ 1,018.00
Fringe	\$ 15,905.53	100%	\$ 15,905.53
Personnel	<u>\$ 42,077.53</u>		<u>\$ 42,077.53</u>
Maint/repair	\$ 4,580.64	100%	\$ 4,580.64
Fuel	\$ 4,115.08	100%	\$ 4,115.08
Tires/Tubes	\$ 853.11	100%	\$ 853.11
Bus Insure	\$ 634.33	100%	\$ 634.33
Operations	<u>\$ 10,183.17</u>		<u>\$ 10,183.16</u>
Operational cost for an additional bus			<u>\$ 52,260.69</u>

Reduced tier savings

District Brecksville Broadview

Item	Annual Avg Cost/bus	Reality adjustment	savings
Reg Bus driver	25,154	0%	\$ -
Sub bus driver	1,018	0%	\$ -
Fringe	\$ 15,905.53	0%	\$ -
Personnel	<u>\$ 42,078.08</u>		<u>\$ -</u>
Maint/repair	\$ 4,580.64	20%	\$ 916.13
Fuel	\$ 4,115.08	20%	\$ 823.02
Tires/Tubes	\$ 853.11	20%	\$ 170.62
Bus Insure	\$ 634.33	0%	\$ -
Operations	<u>\$ 10,183.17</u>		<u>\$ 1,909.77</u>
Cost per bus	<u>\$ 52,261.25</u>	savings	<u>\$ 1,909.77</u>

Added tier cost

District Brecksville Broadview

Item	Annual Avg Cost/bus	Reality adjustment	Cost
Reg Bus driver	25,154	0%	\$ -
Sub bus driver	1,018	0%	\$ -
Fringe	\$ 15,905.53	0%	\$ -
Personnel	<u>\$ 42,078.08</u>		<u>\$ -</u>
Maint/repair	\$ 4,580.64	20%	\$ 916.13
Fuel	\$ 4,115.08	20%	\$ 823.02
Tires/Tubes	\$ 853.11	20%	\$ 170.62
Bus Insure	\$ 634.33	0%	\$ -
Operations	<u>\$ 10,183.17</u>		<u>\$ 1,909.77</u>
Cost per bus	<u>\$ 52,261.25</u>	cost	<u>\$ 1,909.77</u>

Cost basis						
Cost saved by removal of a bus from a tier:				1909		
Cost incurred by adding a bus to a tier:				1909		
Cost to add a bus to the fleet:				52,260		
Scenario 1	Remove tier 3, keep bus numbers constant, no bell time changes					
	Tier 2	add 0 buses				
	Tier 3	delete 20 buses				
	Tier 4	add 0 buses				
	Net impact of 20 tier trips removed			\$ 38,180	savings	
Scenario 2	Remove tier 3, add appropriate buses, no bell time changes					
	Tier 2	add 2 buses				
	Tier 3	delete 20 buses				
	Tier 4	Add 3 buses				
	Net impact of 15 tier trips removed			\$ 28,635	savings	
Scenario 3	Remove tier 3, add 6 buses to MS and Elem, no bell time change					
	Tier 2	add 6 buses, which is 2 tiers and 4 additional buses				
	Tier 3	delete 20 buses				
	Tier 4	add 6 buses				
	Net impact of 12 tier trips removed			22,908	savings	
	Need to add 4 fleet buses			209,040	new cost	
				186,132	Net cost to district	